Measure-Aware Ram Representations and Spectral Certificates for Critical FK–Ising Planar Maps

A dependency audit and conditional constructive analysis of M376

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Abstract

The October 5, 2026 OpenAI preprint on spectral convergence for critical FK–Ising planar maps claims a joint metric, eigenvalue, and heat-trace limit on the ordinary unit-area $\sqrt{3}$-quantum sphere, conditional on its stated companion and Brownian/LQG inputs. We read its complete substantive LaTeX source and audit the chain from electrical estimates through singular-measure traces to Hilbert–Schmidt inverse convergence. This report is not a global verification of that preprint or its companions. We construct an exact Ram displacement representation which transports the actual quantum measure, derive physical finite-coordinate and omitted-space identities, and prove conditional finite approximations without assuming a Lebesgue density or differentiable eigenfunctions. A capped logarithmic kernel gives an explicit Hilbert–Schmidt certificate, with the conservative rate $h^{1/113}$ under the source's ball exponent. A positive inverse-operator estimate turns such certificates into heat-trace bounds uniformly on $[\delta, \infty)$ for every $\delta > 0$. These are locally proved analytic consequences, not a rate in the number of FK edges and not a new solution of the probabilistic scaling-limit problem. Exact and numerical checks, read scope, public literature, and artifact provenance are separately recorded.

Status and confidentiality. Private research report. No private source text was submitted to web searches. The public source pin is

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The pin is supplied by the local catalogue and snapshot, not authenticated by a local Git checkout. Public web retrieval of the pinned directory failed; the locally preserved source is the manuscript reviewed here.

Contents

1 Scope, status, and principal conclusions **4**
1.1 What is established locally 4

2 The model and the normalization contract **4**

3 Source chain and external dependencies **5**
3.1 Public literature inspected 6

4 Audit of locality, electrical control, and conductivity **6**
4.1 What numerical locality actually needs 6
4.2 The fixed-test electrical argument 7
4.3 Conductivity is more than a coordinate change 7

5 Four-point limits and the all-level bottleneck **8**
5.1 Voltage identification away from the poles 8
5.2 Why fixed annuli are insufficient 8

6 Small-ball bounds and logarithmic control **8**

7 Sobolev trace for the singular measure **9**

8 The closed form, inverse, and domain distinction **11**

9 Common Hilbert space and energy compactness **12**

10 Four-point observations and inverse compactness **12**
10.1 A concentration diagnostic 13

11 Exact Ram localization with a transported measure **14**
11.1 What not to infer from localization 14

12 Physical coordinates and full residuals **14**

13 Two convergent finite constructions **15**
13.1 Smooth form trial spaces 15
13.2 Compact inverse compression 16

14 A constructive logarithmic-kernel certificate **16**
14.1 The actual finite matrix 17
14.2 Rate and its honest interpretation 18

15 Positive HS inverses control heat traces **18**

16 Eigenvalue enclosures, padding, and an assembled certificate **19**

17 Stepwise source and private-corpus correspondence **20**

17.1 Attribution of the local additions 21

18 Checks and evidence **21**

18.1 What the checks intentionally do not claim 22

19 Advantages, limitations, and future work **22**

19.1 Advantages within proved scope 22

19.2 Costs and limitations 22

19.3 A carefully separated Weyl-law comparison 23

20 Read scope, provenance, and reproducibility **23**

20.1 Compiler and visual audit contract 24

21 Concluding assessment **24**

1 Scope, status, and principal conclusions

The object of study is manuscript M376, *Spectral convergence for critical FK–Ising planar maps*, attributed to OpenAI and dated October 5, 2026 [1]. The local applicability catalogue assigns level 5, meaning direct core relevance of the Ram operator framework. That rating is an applicability judgment, not a proof certificate. Its prior screening was explicitly bounded; the present author has read all nine substantive source sections, the driver, bibliography, figure source, and manuscript README. Exact file and passage scope appears in Section 20.

The repository README describes manuscripts at different verification stages and acknowledges that unformalized work may have issues. No claim of global verification, peer review of M376, or completed formalization is made here. In particular, the conformal companion C and metric-measure companion P are additional preprint dependencies. We inspect their uses in M376, not every proof in those manuscripts. M376 specifies the October 3, 2026 revision of P.

Three classes of statements are kept distinct throughout:

1. **Source claims and imported inputs:** probabilistic coupling, protected local laws, quantitative address estimates, and the FK spectral limit as a whole.
2. **Deterministic consequences reconstructed here:** trace and inverse construction under explicit ball bounds, abstract inverse compactness, metric-aware coordinates, and spectral readouts.
3. **Local extensions:** measure-transported Ram formulas, a generator-domain warning, a positive capped-kernel certificate, and a quantitative inverse-to-heat estimate. Their proofs are included, with no claim of first publication.

The report's central recommendation is mathematical rather than rhetorical: work with the positive bounded centered inverse K on the physical $L^2(\mu)$ space. Retain its constant-mode convention and transport its measure whenever coordinates change. Approximate K in a norm strong enough to control its spectral mass, then transfer that error to readouts. A finite moment matrix without an omitted-space bound does not do this.

1.1 What is established locally

Under stated hypotheses we obtain an exact singular-measure Ram integral, physical Gram symmetry, a complete residual account, and two constructive approaches to approximation. Smooth form-core dictionaries give convergent variational spaces, while capped Green kernels and mass-weighted cell matrices give explicit inverse error bounds. Positive-part projection is proved harmless in Hilbert–Schmidt distance to the true positive inverse. The final heat bound is uniform even for unbounded time intervals bounded away from zero. None of these steps proves C, P, conductivity homogenization, or random-map compactness from scratch.

2 The model and the normalization contract

Let M_n be the connected spherical planar multigraph with n original edges, decorated by a critical FK configuration A_n. All original edges are present in the walk, not only FK-open edges. If $k(A)$ counts components including isolated vertices, the source's weight is proportional to $2^{\ell(M,A)/2}$, with

$$\ell(M, A) = k(A) + k(A^*) - 1 = 2k(A) + |A| - |V(M)|.$$

Loops and parallel edges are allowed. Each loop contributes two to degree and zero to the squared voltage energy. Put

$$\mu_n(v) = \frac{d_v}{2n}, \qquad D_n = \operatorname{diag}(\mu_n), \qquad Q_n = \sum_{e \text{ nonloop}} b_e b_e^{\mathsf{T}}, \tag{2.1}$$

where b_e is an oriented incidence vector and each parallel edge is a separate term. The walk makes attempts at total rate one, choosing uniformly among the incident half-edges. A loop attempt leaves its position unchanged. Its generator is $L_n = -\operatorname{diag}(d_v)^{-1} Q_n$.

Proposition 2.1 (Exact raw and Brownian normalizations)**.** *On $L^2(\mu_n)$,*

$$B_n = -2nL_n = D_n^{-1} Q_n, \quad \langle f, B_n g\rangle_{\mu_n} = \sum_e d_e f \, d_e g = \mathcal{E}_n^0(f,g), \quad A_n = B_n/2 = -nL_n. \tag{2.2}$$

The kernel is precisely the constants. Thus $\nu_{n,j} = 2n\lambda_{n,j}$ are raw eigenvalues, whereas the accelerated walk eigenvalues are $\nu_{n,j}/2 = n\lambda_{n,j}$.

Proof. For an unoriented nonloop edge $\{u, v\}$, its two half-edge contributions to $-\langle f, L_n g\rangle_{\mu_n}$ sum to $(f(u)-f(v))(g(u)-g(v))/(2n)$. Loop differences vanish. Summation proves the identities. Vanishing energy forces equality across every connecting nonloop edge, hence constancy by connectedness. □

The limiting raw energy is

$$\mathcal{E}^0(f,g) = \int_{\mathbb{S}^2} \nabla f \cdot \nabla g \, \mathrm{dvol}_0, \qquad \mathcal{E}_h = \tfrac{1}{2}\mathcal{E}^0, \quad B_h = 2A_h. \tag{2.3}$$

The form is on $L^2(\mu_h)$; its energy integral uses round area, but its state norm uses quantum area. These two measures cannot be interchanged. Conductivity one for the raw energy, if proved, gives precisely the edge count acceleration n. Changing to successful-jump rate, deleting loop degrees, or using uniform vertex mass changes the operator.

For a concrete check, take two vertices joined by two parallel edges and put one loop at the first vertex. Then $n = 3$, $d = (4, 2)$, $\mu = (2/3, 1/3)$, and

$$Q = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}, \quad -L = \begin{pmatrix} 1/2 & -1/2 \\ -1 & 1 \end{pmatrix}.$$

The nonzero eigenvalues of $-L$, B, and A are respectively $3/2$, 9, and $9/2$. The loop affects the clock and mass even though it contributes no energy term.

3 Source chain and external dependencies

The driver reads the sections in the order 01, 02, 03, 04, 06, 07, 05, 08, 09. This order is substantive: the all-scale compactness section uses the fixed-test electrical bounds, while the continuum trace construction is logically separate from the graph homogenization.

Source	Deliverable used later	Unresolved input or audit boundary
01 introduction	Joint GHP, ordered eigenvalue, positive-time heat limit	Claim of the preprint; not an independently established theorem here.
02 model	Corner isometry, flag geometry, common coordinate and two tours	C: encoding, topology, uniformization; P: metric and local field comparison.
03 locality	Numerical observation kernels, support transfer, deterministic germs	Protected raw-word resampling and conditional independence from C/P.
04 network	Annular lower bounds, cutoffs, circuit traffic, route laws	Aligned-annulus theorem of C; exact primal-dual drawing and local arrays.
06 conductivity	Uniform-topology relaxed energy is the raw Dirichlet integral	Fixed-test bounds, germ determinism, square boundary approximation.
07 Green	Unit-current four-point voltage limit at distinct points	Local energy recovery and harmonic compactness; not a diagonal kernel bound.
05 compactness	One threshold for all levels and all functions	P: address charge, finite-view construction, filling and support transfer.
08 continuum	Trace, full quasi-support, centered HS inverse, heat finiteness	GMC moment input and sphere local absolute continuity; deterministic proof follows.
09 spectrum	Common-space compactness, HS inverse convergence, spectral limit	Both four-point identification and all-level compactness, not either alone.

3.1 Public literature inspected

Current primary public pages were consulted on October 7, 2026. The mated-CRT walk theorem of Berestycki–Gwynne [7] concerns a different map family and uses a median-exit time scale; Remark 1.3 bounds that scale up to constants and leaves its limiting multiplicative constant unresolved in that work. It does not by itself prove M376's exact FK clock. Maillard–Rhodes–Vargas–Zeitouni [8], Section 3, supplies the precedent of a centered Hilbert–Schmidt Green operator in a torus setting. Andres–Kajino [9] studies Liouville heat-kernel regularity and spectral dimensions. Robert–Vargas [10] supplies GMC moment context. These works support the analytic setting, not the complete FK dependency chain.

Chen–Yu's September 2026 preprint [14] gives a discrete approximation scheme for time-changed Brownian motion under smooth-measure/full-quasi-support hypotheses. It is not a statement about the particular random FK graph sequence. The $\sqrt{8/3}$ heat-kernel work [15] has a different parameter from $\sqrt{3}$. We do not transfer its parameter-specific bounds.

4 Audit of locality, electrical control, and conductivity

4.1 What numerical locality actually needs

Source `03-locality.tex`, `prop:readout-extension`, retains arbitrary measurable compactified numerical outputs of a finite protected incidence record. Continuity of resistance as a function of limiting contours is not assumed. The proof instead pushes the exact raw-word weighted law through the same observation in actual and reference experiments. The limiting weight must depend

only on retained continuum inputs, not on the new numerical output. Complete stars, exact gluing, strict spatial buffers, and positive effective units are part of the contract.

The spatial-selection step also requires care: an observation chosen from the extracted local atlas is evaluated through its intrinsic description. The global chart label need not be locally measurable. Conformal comparison maps whose conditional laws depend on the output array are kept outside the conditioning field. These safeguards are consequential, because merely asserting "the object is local" would not give a product kernel.

`prop:sphere-support` transfers null events, with actual additive field height and unchanged units. It supplies eventual pathwise bounds in a representation extraction, not finite-scale probabilities. Likewise the selected-slot entropy estimate preserves compatible subprobabilities without division by their potentially small mass. All these assertions still depend on the stated companion constructions.

4.2 The fixed-test electrical argument

Source `04-network.tex` first identifies edge extremal length with the minimum squared expected traffic. Its convex-hull proof is finite-dimensional: the least-norm point p in the closed hull of path-usage vectors satisfies $a_\pi \cdot p \geq \|p\|^2$. Testing the extremal-length supremum at p gives equality. For port-to-port paths this is wired resistance in the same confinement graph. A lower bound transfers to a larger family only when every larger path contains a tested subpath.

The argument against collapsing raw resistance uses record-low quantiles, aligned shells, a density budget, and tight path laws. It then obtains rectifiable limiting paths with an average arclength measure absolutely continuous with planar area. Local germ determinism and rotational covariance spread cheap crossings to transverse primal and dual families. Their forced intersection yields, for expected traffics τ, τ^*,

$$1 \leq \sum_e \tau(e)\tau^*(e^*) \leq \|\tau\|_2 \|\tau^*\|_2 \longrightarrow 0,$$

a contradiction. The independence here is only the product of two fixed path laws after the entire sample is fixed; the primal and dual maps are not asserted independent.

4.3 Conductivity is more than a coordinate change

Source `06-conductivity.tex` builds an inner-domain, tolerance-relaxed energy for locally uniform convergence. Countably many finite quadratic infima yield a sandwich valid for all continuous test functions. Cutoff gluing produces local quadratic energy measures, and circuit comparisons bound them by Sobolev energy. Polarization and rational affine gradients identify a tensor density. Local germ determinism, translation covariance, and rotation covariance reduce it to a positive deterministic scalar. Only then do primal-dual equality and complementary square conductances force $k_{\mathrm{p}} k_{\mathrm{d}} = 1$ and $k_{\mathrm{p}} = k_{\mathrm{d}}$, hence both equal one.

This is not uniform ellipticity of a quantum-density generator. It is an identification of the conformal energy integral. Nor is it already Mosco convergence on varying L^2 spaces. The source explicitly postpones the small-mass concentration problem. This report preserves both distinctions.

5 Four-point limits and the all-level bottleneck

5.1 Voltage identification away from the poles

For $Q_n u_n = \delta_a - \delta_b$, the raw variational functional is $\mathcal{E}_n^0(u) - 2(u(a) - u(b))$. Source `07-green.tex` clips voltages to obtain $\mathcal{E}_n^0(Tu_n) \le \beta - \alpha$, combines this with circuit traffic, and bounds oscillations away from the poles. Local replacement by energy recoveries rules out a limiting energy defect and identifies harmonicity. The source uses continuity sets of the vague energy measure, which avoids using the open-set Portmanteau inequality in the wrong direction.

Varying the source voltage by a cutoff identifies flux $+1$ at the source and -1 at the sink. One-sided bounds and the isolated harmonic-singularity argument exclude higher-order poles. The limiting voltage is therefore

$$U_{x,y;z_0}(z) = \frac{1}{2\pi} \log \frac{\operatorname{ch}(z,y)\operatorname{ch}(z_0,x)}{\operatorname{ch}(z,x)\operatorname{ch}(z_0,y)}. \tag{5.1}$$

This concerns four distinct limiting positions. It does not give an integrable bound on a neighborhood of the diagonal.

5.2 Why fixed annuli are insufficient

Source `05-compactness.tex` is the critical additional module. The fixed-test thresholds of the electrical theorem depend on the test. Spectral compactness needs a single eventual threshold controlling every finer dyadic scale down to singleton cuts and every vertex function.

The source uses P's address estimate with $\vartheta = 7/6$. Occupation entropy is outweighed by monochrome rank charge. A one-base rank-credit lemma allows a wide log-scale range; a separate many-base input handles a narrow range. Truncated packing is proved with the probe stride held fixed before its later limit, rather than exchanging those limits. Off-center finite-view recipes yield surrounding circuits with controlled squared traffic, restricted support reach, and a bound on uninterrupted tour runs inside a disk. The selected anchor law is compared as a subprobability, even when anchors depend on the observed output.

The overlap bound is $Ce^{N/40}$. Circuit boundary oscillation has a bill $e^{N/10}\sqrt{\mathcal{E}}$; child mass is $O(e^{-N})$. The exact exponent ledger is

$$-1 + \frac{1}{5} + \frac{1}{40} = -\frac{31}{40}, \qquad \frac{31}{40} > p = \frac{7}{10} > \frac{1}{2}. \tag{5.2}$$

Telescoping square-root errors to singleton blocks and then using the orthogonal block average gives the announced exponent p. Circuit choices may depend on the function, but the support sets and overlap event may not. Finally the smallest sufficient all-level threshold is itself a retained finite-record numerical output. Tightness is established before support transfer to sphere patches. This is the source's claimed route to the necessary quantifier order; the Ram reformulation does not replace it.

6 Small-ball bounds and logarithmic control

Here and below the analysis is deterministic unless explicitly labelled as an input about the quantum sphere. Write d_0 for round distance, $\sigma_0 = \mathrm{vol}_0/(4\pi)$, and assume a probability measure μ obeys

$$cr^b \le \mu(B_{d_0}(x,r)) \le Cr^\alpha \quad (x \in \mathbb{S}^2,\ 0 < r \le r_*), \qquad \alpha > 0, \tag{6.1}$$

with finite positive constants. M376 claims $\alpha = 1/56$ and $b = 8$ almost surely for its ordinary unit-area sphere, with random constants.

The source's GMC calculation is checkable at the exponent level. For $\gamma^2 = 3$, $\zeta(q) = \frac{7}{2}q - \frac{3}{2}q^2$, and

$$\zeta(7/6) = 49/24, \quad \zeta(7/6) - (7/6)(1/56) = 97/48, \quad \zeta(-1) = -5.$$

A two-dimensional grid costs r^{-2}, leaving summable dyadic probabilities $O(r^{1/48})$ and $O(r)$ for the upper and lower bounds. Local absolute continuity modulo constants and two disjoint marking triples then provide a finite chart cover of the sphere. This transfer is a genuine sphere-law input, not implied by the moment arithmetic alone.

Lemma 6.1 (Logarithmic moments)**.** *The upper bound in* (6.1) *implies nonatomicity and, for every* $q > 0$*,*

$$\sup_x \int \log_+\left(\frac{r_*}{d_0(x,y)}\right)^q \mathrm{d}\mu(y) \le C r_*^\alpha \frac{\Gamma(q+1)}{\alpha^q}. \tag{6.2}$$

In particular the round Green kernel is in $L^2(\mu \otimes \mu)$*.*

Proof. Shrinking a ball to a point gives zero mass. Layer cake and Tonelli give

$$q \int_0^\infty t^{q-1} \mu(B(x, r_* e^{-t}))\, \mathrm{d}t \le C q r_*^\alpha \int_0^\infty t^{q-1} e^{-\alpha t}\, \mathrm{d}t.$$

The last integral is $\Gamma(q)/\alpha^q$. The round Green kernel equals $-(2\pi)^{-1} \log d_0(x,y)$ plus a bounded function near the diagonal and is smooth away from it. Use $q = 2$ and integrate over x. □

No lower density with respect to round area is present in (6.1). For $b = 8$ the lower bound is much smaller than an area-scale lower bound. It is a support and capacity-control hypothesis, not a uniformly elliptic weight. Similarly, the small positive upper exponent is enough for logarithmic integrability but far too weak to imply efficient high-order quadrature.

7 Sobolev trace for the singular measure

Let G have zero round mean and distributional normalization

$$\mathcal{E}^0(G(\cdot, y), \phi) = \phi(y) - \sigma_0(\phi). \tag{7.1}$$

The kernel is symmetric. We reconstruct the main mechanism of source `lem:trace-green`, retaining the step that proves injectivity.

Theorem 7.1 (Trace and potential)**.** *Under* (6.1)*, smooth restriction extends to a bounded injective map* $R_\mu : H^1(\mathbb{S}^2) \to L^2(\mu)$*, given by quasi-continuous representatives. The measure charges no* H^1*-polar set and*

$$\|R_\mu(f - \sigma_0 f)\|_{L^2(\mu)} \le C_\mu \mathcal{E}^0(f)^{1/2}, \tag{7.2}$$

$$\|f\|_{H^1} \asymp_\mu (\|R_\mu f\|^2_{L^2(\mu)} + \mathcal{E}^0(f))^{1/2}. \tag{7.3}$$

For $g \in L^2(\mu)$ *the round-mean-zero potential* $u_g \in H^1$ *satisfies*

$$\mathcal{E}^0(u_g, f) = \int g R_\mu f\, \mathrm{d}\mu - \mu(g)\sigma_0(f), \qquad R_\mu u_g(x) = \int G(x,y) g(y)\, \mathrm{d}\mu(y). \tag{7.4}$$

Proof. Put $s_g = g\mu - \mu(g)\sigma_0$. The first logarithmic moment and Schur's inequality bound the absolute logarithmic interaction of $|s_g|$ and $|s_k|$ by $C\|g\|_2\|k\|_2$. Let P_t be the round heat semigroup with generator Δ_0, and smooth these signed sources. If $u_{g,t}$ is the zero-round-mean solution for $P_t^* s_g$, integration by parts gives

$$\mathcal{E}^0(u_{g,t}, u_{k,s}) = \iint G_{t+s}(x,y)\,\mathrm{d}s_g(x)\,\mathrm{d}s_k(y), \quad G_t = \int_t^\infty (p_v - (4\pi)^{-1})\,\mathrm{d}v.$$

The smooth sphere heat kernel bounds $|G_t|$ by a fixed multiple of $1 + \log_+(r_*/d_0)$ off the diagonal; $G_t \to G$ there. The product measures have no diagonal atoms. Dominated convergence applied to the three cross-energy terms makes $u_{g,t}$ Cauchy in energy and, by round Poincare, in H^1. Its limit has energy at most $C\|g\|_2^2$ and the weak equation for smooth tests. Duality against all g proves (7.2) for smooth f, then extends restriction by density.

For compact F and smooth $\phi \geq 1$ near F, the restriction bound gives $\mu(F) \leq C\|\phi\|_{H^1}^2$. Infimizing proves capacity domination, and regularity extends polar-nullity to Borel sets. A rapidly converging smooth approximation to f converges quasi-everywhere after a subsequence; another subsequence converges in μ-almost-everywhere trace values. Polar-nullity identifies these representatives. The weak equation therefore extends to all H^1 tests.

Injectivity needs more. Work in a buffered Euclidean chart and normalize $\mu|_{B_r(x)}$ to a probability, then rescale to B_1. Call the result $\widehat{\mu}_{x,r}$. Both ball bounds imply

$$\widehat{\mu}_{x,r}(B_s(y)) \leq \min\{1, A_r s^\alpha\}, \quad A_r = \max(1, C_1 r^{\alpha-b}).$$

Integrating this upper bound against logarithmic layer cake gives $\sup_y \int \log_+(1/|y-z|)\,\mathrm{d}\widehat{\mu}_{x,r}(z) \leq C(1 + |\log r|)$. The Dirichlet Green kernel on B_3, restricted to $B_1 \times B_1$, has the same logarithmic upper bound. The smoothing argument above, with zero boundary condition, gives

$$\|\widehat{\mu}_{x,r} - |B_1|^{-1}\mathbf{1}_{B_1}\,\mathrm{d}z\|_{H^{-1}(B_3)} \leq C\sqrt{1 + |\log r|}.$$

Test on a cutoff times $f(x + r\,\cdot)$ minus its B_3 average. Poincare controls the cutoff error and two-dimensional energy is invariant under this scaling. Consequently

$$\left|\frac{1}{|B_r|}\int_{B_r(x)} f\,\mathrm{d}z - \frac{\int_{B_r(x)} R_\mu f\,\mathrm{d}\mu}{\mu(B_r(x))}\right| \leq C\sqrt{1 + |\log r|}\left(\int_{B_{3r}(x)} |\nabla f|^2\,\mathrm{d}z\right)^{1/2}. \tag{7.5}$$

Smooth approximation extends this inequality to H^1. If the trace is zero, at almost every common Lebesgue point of f and $|\nabla f|^2$ the right side is $O_x(r\sqrt{1 + |\log r|}) \to 0$. Hence $f = 0$ round-almost everywhere. This proves injectivity.

Write $f = c + v$ with $\sigma_0(v) = 0$. Round Poincare controls v in H^1 by energy, and $|c| \leq \|R_\mu f\|_2 + C_\mu \mathcal{E}^0(f)^{1/2}$. This proves (7.3). Finally pair $R_\mu u_g$ against k. The cross-energy formula and zero round means remove the σ_0 terms, giving $\int k R_\mu u_g\,\mathrm{d}\mu = \iint G(x,y)k(x)g(y)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y)$. The second logarithmic moment makes this an L^2 kernel identity. □

The conclusion is an injective bounded trace, not an identification of $L^2(\mu)$ with $L^2(\mathrm{vol}_0)$. In particular a spatially smooth coefficient formula cannot justify replacing quantum mass by area quadrature.

8 The closed form, inverse, and domain distinction

Let $P_0 = I - \Pi_\mu$, where $\Pi_\mu g = \mu(g)\mathbf{1}$. Define

$$G_\mu(x,y) = G(x,y) - \int G(x,z)\,\mathrm{d}\mu(z) - \int G(z,y)\,\mathrm{d}\mu(z) + \iint G(z,w)\,\mathrm{d}\mu(z)\,\mathrm{d}\mu(w), \qquad K = P_0 T_G P_0. \tag{8.1}$$

The constant mode is killed by K and must later be added once to the heat trace. A constant in the original Green kernel cancels from this formula.

Proposition 8.1 (Closed form and centered inverse)**.** *Under* (6.1)*, the closure of the smooth half-energy has domain $R_\mu H^1$, is a regular Dirichlet form, and has operator $A = B/2$. On $L^2_0(\mu)$, K is the positive injective inverse of B. It is Hilbert–Schmidt and A has compact resolvent and finite heat trace for all positive times.*

Proof. Injectivity defines the transported form uniquely, and (7.3) makes its domain complete. Smooth functions are dense in H^1, and their restrictions are dense in $L^2(\mu)$ by uniform density in continuous functions. Normal contractions decrease the energy and commute with quasi-continuous trace. This proves closedness, the Markov property, and regularity with the smooth core. On centered traces, (7.2) gives $\|v\|_2 \le 2C_\mu \mathcal{E}^0(v)^{1/2}$, interpreting energy through the unique preimage. For centered g, (7.4) shows that $Kg = R_\mu u_g - \mu(R_\mu u_g)$ has weak energy pairing $\langle g, f\rangle$. Thus $BKg = g$, with uniqueness from centered coercivity. Positivity follows from $\langle g, Kg\rangle = \mathcal{E}^0(u_g) \ge 0$.

Double centering is an orthogonal projection on the kernel space, so K is HS. Also $(B+I)^{-1} = \Pi_\mu + K(I+K)^{-1}$ is compact. Its only zero-energy mode is constant. If $\kappa_i = 1/\nu_i$ are positive inverse eigenvalues, then $\sum_i \nu_i^{-2} = \|K\|^2_{\mathrm{HS}} < \infty$. The bound $e^{-ts/2} \le 16 e^{-2} t^{-2} s^{-2}$ proves heat finiteness and local uniform convergence of the series. □

Identification with Liouville Brownian motion additionally uses the usual time-change theorem for smooth measures of full quasi-support [11]. The trace theorem gives polar-nullity and injectivity; these are the relevant inputs. The underlying Brownian generator is $\Delta_0/2$.

Proposition 8.2 (Smooth form core is not a generator core)**.** *Assume additionally $\mu \perp \mathrm{vol}_0$. If $f \in C^\infty(\mathbb{S}^2)$ and $R_\mu f \in \mathrm{Dom}(B)$, then f is constant.*

Proof. For some $g \in L^2(\mu)$ the generator identity tested against all smooth ϕ reads $\int(-\Delta_0 f)\phi\,\mathrm{dvol}_0 = \int g\phi\,\mathrm{d}\mu$. Both sides are finite signed measures, so equality of smooth tests gives equality of measures. Their mutual singularity forces each to be zero. Thus $\Delta_0 f = 0$, and integration by parts on the sphere implies $\int |\nabla f|^2 = 0$. □

This conditional statement explains a concrete hazard in importing a generator residual formula from the private corpus. Smooth dictionaries are valid *form* trial spaces. They cannot automatically be inserted in $\|(B-\lambda)f\|_2$. Bounded inverse compression avoids that domain problem, and Green images Kg do belong to $\mathrm{Dom}(B)$.

9 Common Hilbert space and energy compactness

The source represents the $2n$ corner occurrences in tour order as equal intervals of $[0,1]$. A vertex v occurs d_v times, so

$$U_n f(t) = f(v_n(\lfloor 2nt \rfloor)), \qquad \|U_n f\|^2_{L^2[0,1]} = \sum_v \mu_n(v)|f(v)|^2. \tag{9.1}$$

The image $\mathcal{H}_n$ is generally a proper subspace. Extending an inverse by zero to $\mathcal{H}_n^\perp$ does not add physical zero modes to the graph generator. The continuum tour has μ-almost-everywhere singleton fibers by the source's geometric input, making $Uf = f \circ \eta$ a unitary map onto $L^2[0,1]$. Pushforward alone would only prove an isometry, not surjectivity.

Two rooted tours are matched by a measure-preserving permutation of corner intervals. Uniform tracking and singleton fibers identify its almost-everywhere limit. This ensures convergence of the finitely many block indicators at a fixed dyadic level. In the priority cover, coefficients are averages over the original positive-length blocks, not over possibly vanishing priority pieces. That distinction keeps the coefficients uniformly bounded.

Theorem 9.1 (Finite-rank approximation consequences)**.** *Let B_n be positive on centered finite-dimensional subspaces $\mathcal{H}_n$ of a fixed Hilbert space H, with constants as its only zero mode. Suppose for all $n \ge n_0, j \ge J$ there are linear maps $Q_{n,j} : \mathcal{H}_n \to H$ with*

$$\operatorname{rank} Q_{n,j} \le C_0 2^j, \qquad \|F - Q_{n,j}F\|^2 \le C_1 2^{-pj}\langle F, B_n F\rangle, \qquad p > 0. \tag{9.2}$$

Assume also that, for each fixed j, images of every norm-bounded sequence $F_n \in \mathcal{H}_n$ have a strongly convergent subsequence. Then bounded norm and energy imply subsequential strong compactness. Moreover, for sufficiently large i, with infinite padding,

$$\nu_{n,i} \ge c_* i^p, \qquad c_* = C_1^{-1}(4C_0)^{-p}. \tag{9.3}$$

Proof. Diagonalize the fixed-level compact extractions over j. For a given $\epsilon > 0$ choose j making the square root of (9.2) less than $\epsilon/3$ uniformly on the energy-bounded sequence. Two late level-j approximants differ by less than $\epsilon/3$, so the original sequence is Cauchy along this diagonal. For the eigenvalue bound put $j = \lfloor \log_2(i/(2C_0)) \rfloor \ge J$. The span of the first $i+1$ eigenvectors, including the constant, contains a unit vector killed by $Q_{n,j}$ because its rank is at most $i/2$. Its energy is at most $\nu_{n,i}$. Thus $1 \le C_1 2^{-pj}\nu_{n,i}$, giving the stated constant. Padded eigenvalues are $+\infty$ and obey the same bound. □

The fixed-level compactness assumption is essential: a rank bound alone allows a rank-one range to rotate through an orthonormal sequence. In M376 it is furnished by the tour matching and convergent block indicators. The resulting constants and thresholds are sample-dependent but simultaneous in all later n, all $j \ge J$, and all functions, on the selected extraction.

10 Four-point observations and inverse compactness

For a probability space (X, μ) define

$$(\mathfrak{D}k)(x, x', y, y') = k(x, y) - k(x, y') - k(x', y) + k(x', y').$$

Call a kernel doubly centered if both marginal integrals vanish.

Lemma 10.1 (Exact four-point norm)**.** *For doubly centered $k \in L^2(\mu^2)$,*

$$k(x,y) = \iint \mathfrak{D}k(x,x',y,y')\,\mathrm{d}\mu(x')\,\mathrm{d}\mu(y'), \qquad \|\mathfrak{D}k\|_{L^2(\mu^4)} = 2\|k\|_{L^2(\mu^2)}. \tag{10.1}$$

Proof. Integrate the four terms. Three vanish by centering. In the squared norm expansion the four square terms each give $\|k\|_2^2$ and every cross term vanishes after integrating an unshared variable. These operations are justified by Fubini and Cauchy–Schwarz. □

Theorem 10.2 (Source inverse criterion, reconstructed)**.** *Suppose $H = L^2[0,1]$, the energy-bounded sequences of Theorem 9.1 are strongly precompact, and (9.3) holds with $p > 1/2$. Let K_n be the centered inverse of B_n, zero on $\mathcal{H}_n^{\perp}$, with kernel k_n. Suppose a doubly centered $k \in L^2([0,1]^2)$ satisfies $\mathfrak{D}k_n \to \mathfrak{D}k$ almost everywhere. Then $\|K_n - K\|_{\mathrm{HS}} = \|k_n - k\|_2 \to 0$.*

Proof. List $\kappa_{n,i} = 1/\nu_{n,i}$ decreasingly with zero padding. Uniformly for all large n,

$$\sum_{i>N} \kappa_{n,i}^2 \le c_*^{-2} \sum_{i>N} i^{-2p} \le \frac{c_*^{-2}}{2p-1} N^{1-2p}. \tag{10.2}$$

First bound the largest inverse eigenvalue. If $M_n = \|K_n\|_{\mathrm{op}}$ tends to infinity on a subsequence, extract each normalized eigenvalue $\alpha_{n,i} = \kappa_{n,i}/M_n \to \alpha_i$. If $\alpha_i > 0$, its unit eigenvector has energy $(M_n\alpha_{n,i})^{-1} \to 0$ and therefore a strong subsequential limit. If $\alpha_i = 0$, its rank-one summand vanishes in HS norm regardless of its eigenvector. The tail (10.2) also controls the normalized sequence for $M_n \ge 1$. Finite truncations and the uniform tail thus give an HS limit K^* of norm one for K_n/M_n. But its four-point kernel converges almost everywhere to zero, while HS convergence makes the same kernels converge in L^2 to $\mathfrak{D}k^*$. Uniqueness of limits in measure gives $\mathfrak{D}k^* = 0$, and centering with Lemma 10.1 gives $K^* = 0$, a contradiction.

Now the $\kappa_{n,i}$ are bounded. On any subsequence extract their coordinatewise limits. Each positive limiting coefficient has eigenvectors of bounded energy and hence a strongly convergent further subsequence; zero coefficients again cost nothing. The same HS tail yields a kernel limit k^*. Almost-everywhere four-point identification gives $\mathfrak{D}k^* = \mathfrak{D}k$, so $k^* = k$. Every subsequence has a further subsequence with this same limit, proving convergence. □

For the graph kernel relative to μ_n, $Q_n k_n(\cdot, y) = \delta_y - \mu_n$, so a difference of columns is a unit-current voltage. Equation (5.1) therefore identifies $\mathfrak{D}k_n$ after tour pullback. Nonatomicity makes almost every quadruple have distinct limiting positions. This explains why the voltage theorem suffices *together with* compactness and the HS tail. It does not assert dominated convergence directly from pointwise voltages.

10.1 A concentration diagnostic

Choose disjoint sets $A_n \subset [0,1]$ with masses $m_n \to 0$, and let $e_n = (\mathbf{1}_{A_n} - m_n)/\sqrt{m_n(1-m_n)}$. These are centered unit vectors with no strongly convergent subsequence. The kernels $k_n = e_n \otimes e_n$ have HS norm one and centered rank one. For almost every quadruple, each coordinate eventually avoids A_n, and $\mathfrak{D}k_n = 0$ eventually. Thus four-point convergence almost everywhere does not imply HS convergence. Defining B_n to be one on e_n and zero on constants gives an example where the padded high-index bound is vacuous but energy compactness fails. This is a positive-operator diagnostic, separate from the indefinite spectral-pollution example in the private corpus.

11 Exact Ram localization with a transported measure

The private foundational report [4] defines $h(s,\alpha) = h'(s+\alpha, s)$ and $h'(x,s) = h(s, x-s)$. Its global/Ram equivalence is an exact change of variables; IRLT inverts the kernel description, not the integral operator. Here the relevant integral is against a singular measure, so the measure must undergo the same change of variables.

Theorem 11.1 (Measure-aware Ram identity)**.** *Let $V \subset \mathbb{R}^d$ be Borel, let ρ be a finite Borel measure, and let $Tf(x) = \int_V H(x,y) f(y)\,\mathrm{d}\rho(y)$ be absolutely defined for the relevant x. Set*

$$\rho_x = (y \mapsto x - y)_{\#}\rho, \qquad h(s,\alpha) = H(s+\alpha, s). \tag{11.1}$$

Then exactly

$$Tf(x) = \int_{x-V} h(x-\alpha,\alpha) f(x-\alpha)\,\mathrm{d}\rho_x(\alpha). \tag{11.2}$$

If $H \in L^2(\rho \otimes \rho)$, this defines the same bounded HS operator in either notation, with the same spectrum and norm.

Proof. The definition of pushforward states $\int q(\alpha)\,\mathrm{d}\rho_x(\alpha) = \int q(x-y)\,\mathrm{d}\rho(y)$ for every integrable q. Apply it to the displayed integrand. For an L^2 kernel, Cauchy–Schwarz makes the operator well defined for almost every x and bounded by its kernel norm. Equality is an equality of the same physical operator, so no spectral change occurs. □

For a sphere chart $\phi : U \to V$, put $\rho = \phi_{\#}(\mu|_U)$. Use the kernel $H(x,y) = G_\mu(\phi^{-1}x, \phi^{-1}y)$, with desired bounded cutoffs included in H. The theorem applies even when ρ is singular. In the discrete version ρ_x is the atomic measure $\sum_v \mu_n(v)\delta_{x-z_n(v)}$ in a chart. The weights are corner masses, not equal vertex masses.

For a global implementation choose a finite smooth partition of unity χ_a on sphere charts. Expand the kernel as $G_\mu(x,y) = \sum_{a,b} \chi_a(x) G_\mu(x,y) \chi_b(y)$. Each rectangular chart block can use the observation coordinate of chart a and source coordinate of chart b in (11.1). Both are Euclidean coordinates of the same dimension; $s+\alpha$ is only the observation coordinate, not a claimed intrinsic sphere translation. Summing the finite blocks restores the original operator exactly. Since the cutoffs are bounded, their products with G_μ remain HS.

11.1 What not to infer from localization

Generally ρ_x depends on x. Differentiating (11.2) as though $\rho_x = \mathrm{d}\alpha$ loses the measure dependence. The Green kernel is not supported near zero displacement and has a logarithmic singularity there. The source gives no analytic derivative jet for a typical eigenfunction. Thus the foundational report's Taylor solution scheme is not applied without its remainder, differentiability, domain, and stability hypotheses. We instead use bounded kernels and finite physical spaces for which a remainder is explicitly accounted for.

12 Physical coordinates and full residuals

This section specializes the private physical-metric chapter [5] to $H = L^2_0(\mu)$ and the bounded self-adjoint operator K. Take linearly independent $\psi_1, \ldots, \psi_r \in H$, synthesis $Jc = \sum_j c_j \psi_j$, and

define

$$G_r = J^*J, \quad C_r = J^*KJ, \quad P_r = JG_r^{-1}J^*, \quad T_r = G_r^{-1}C_r. \tag{12.1}$$

Here $G_r > 0$, and P_r is the physical orthogonal projection.

Proposition 12.1 (Metric symmetry and residual account)**.** *One has $G_rT_r = T_r^*G_r$ and $\widehat{T}_r = G_r^{-1/2}C_rG_r^{-1/2}$ is Hermitian and nonnegative. For any $\kappa \in \mathbb{R}$,*

$$\|(K-\kappa)Jc\|^2 = \|(T_r-\kappa)c\|_{G_r}^2 + \|(I-P_r)KJc\|^2. \tag{12.2}$$

Consequently a full residual $\epsilon\|Jc\|$ implies $\operatorname{dist}(\kappa,\sigma(K)) \le \epsilon$ for $Jc \neq 0$.

Proof. Self-adjointness gives $C_r = C_r^*$; positivity follows from that of K. The projected component of $(K-\kappa)Jc$ is $J(T_r-\kappa)c$, and the orthogonal component is $(I-P_r)KJc$. Pythagoras proves the identity. The last assertion is the spectral-theorem inequality $\|(K-\kappa)u\| \ge \operatorname{dist}(\kappa,\sigma(K))\|u\|$. □

A zero finite residual only removes the first term of (12.2). The private spectral chapter [6] gives an explicit indefinite nested graph-core pollution example and explains why strong resolvent convergence is weaker than norm convergence. Its counterexample is not claimed to describe M376's positive compact inverse. It identifies a logical error to avoid: finite algebra is not a uniform continuum estimate.

Proposition 12.2 (Exact HS omission budget)**.** *For any finite-rank orthogonal P and self-adjoint HS K,*

$$\|K-PKP\|_{\mathrm{HS}}^2 = 2\|(I-P)KP\|_{\mathrm{HS}}^2 + \|(I-P)K(I-P)\|_{\mathrm{HS}}^2. \tag{12.3}$$

Proof. Decompose $H = PH \oplus (I-P)H$. The difference has blocks $\left(\begin{smallmatrix} 0 & C^* \\ C & D \end{smallmatrix}\right)$. Sum the squared matrix coefficients in orthonormal bases of the two spaces; the three blocks are orthogonal in the HS inner product. □

This is a concrete completeness requirement for a finite Ram/moment calculation: it must bound both leakage C and the entire complementary block D. Positivity of a finite Gram matrix supplies neither. Relative Gram error is the proper numerical scale: if $aG_r \preceq \widetilde{G}_r \preceq bG_r$, then the corresponding norms are bounded by $\sqrt{a}$ and $\sqrt{b}$ times the physical norm. Entrywise closeness alone cannot control nearly null Gram directions.

13 Two convergent finite constructions

13.1 Smooth form trial spaces

Take real spherical harmonics of positive degree and subtract their μ means. Let X_r be their increasing finite spans after trace. Full support and continuity imply independence: a finite harmonic sum equal to a constant μ-almost everywhere is equal everywhere, and orthogonality under round area then makes all positive-degree coefficients zero. Thus every finite physical Gram matrix is positive definite.

Finite spherical harmonic sums are dense in H^1. The trace norm equivalence makes the centered spans a core for the centered form domain. For the generalized energy matrix

$$M_{ij} = \int \psi_i\psi_j \,\mathrm{d}\mu, \qquad E_{ij} = \int \nabla\psi_i \cdot \nabla\psi_j \,\mathrm{dvol}_0,$$

the generalized eigenvalues of (E, M) are variational upper bounds for the raw eigenvalues, and decrease to each fixed raw eigenvalue as the spaces increase. To see convergence, approximate a basis of the first k continuum eigenfunctions in form norm. The finite mass and energy matrices converge, hence the maximum Rayleigh quotient on that approximating span converges to ν_k. Min–max gives the lower bound ν_k for every trial space and the matching limiting upper bound. Multiplicities are handled by the whole k-dimensional spectral subspace.

These trial functions do not need to be in $\mathrm{Dom}(B)$, and no strong generator residual is asserted. The inverse of the finite energy matrix is a variational solution operator; it is generally not $P_r K P_r$. For the latter the actual entries are $J^* K J$, as in (12.1). The two constructions coincide only under additional invariance.

13.2 Compact inverse compression

If finite-rank orthogonal projections $P_r \to I$ strongly on $L^2_0(\mu)$, then

$$\|K - P_r K P_r\|_{\mathrm{HS}} \longrightarrow 0. \tag{13.1}$$

Indeed, approximate K first by a finite sum of rank-one spectral terms in HS norm. Strong convergence of P_r gives convergence for that finite sum. Contraction of the projections bounds the remaining two errors by the spectral remainder. This proves the claim by an arbitrary-ϵ argument, without a quantitative rate. A nested cell partition generating the Borel sigma-field gives another such family through conditional expectation. Cell indicators are legitimate for bounded K even when they are outside the form domain.

In the complete centered space of a finite graph, an injective synthesis J with range $\mathbf{1}^\perp$ gives

$$E = J^{\mathsf{T}} Q J, \quad G = J^{\mathsf{T}} D_n J, \quad K_n = J E^{-1} J^{\mathsf{T}} D_n, \qquad k_n = J E^{-1} J^{\mathsf{T}}. \tag{13.2}$$

Here k_n is a kernel relative to μ_n. The weak equation proves $B_n K_n = K_n B_n = I - \mathbf{1}\mu_n^{\mathsf{T}}$ and $Q_n k_n = I - \mu_n \mathbf{1}^{\mathsf{T}}$. These formulas depend on completeness of the centered synthesis. Using the same formula on a strict trial subspace only gives a Galerkin inverse and leaves a complementary-space error.

14 A constructive logarithmic-kernel certificate

The following local extension was proposed and independently derived in the parent audit notes; we include the full proof. It needs no Taylor expansion of eigenfunctions and no density of quantum area.

Use chordal distance $d(x, y) = |x - y|_{\mathbb{R}^3}$ on the unit sphere and assume

$$\mu(B_d(x, r)) \le C r^\alpha \quad (0 < r \le r_0), \qquad G(x, y) = -\frac{1}{2\pi} \log d(x, y) + c_0. \tag{14.1}$$

A round-distance bound implies this after changing constants, since $d \le d_0 \le (\pi/2) d$. Assume $K = P_0 T_G P_0 \ge 0$, as follows from Proposition 8.1 for the measure in that proposition.

Theorem 14.1 (Cap, cells, centering, and positivity)**.** *Choose $0 < \epsilon \le r_0$, a finite measurable partition into positive-mass cells C_a of chordal diameter at most h, and representatives $x_a \in C_a$. Set $w_a = \mu(C_a)$ and*

$$G_\epsilon(x, y) = -\frac{1}{2\pi} \log \max\{d(x, y), \epsilon\} + c_0, \quad \widehat{G}(x, y) = G_\epsilon(x_a, x_b) \ (x \in C_a, y \in C_b).$$

Let $\widehat{K} = P_0 T_{\widehat{G}} P_0$ and $S = (\widehat{K})_+$, the positive part in the physical Hilbert space. Then S is positive, centered, of rank at most $q-1$ for q cells, and

$$\|K - S\|_{\mathrm{HS}} \le \eta(h,\epsilon) := \frac{\sqrt{C}}{\sqrt{2}\pi\alpha}\epsilon^{\alpha/2} + \frac{h}{\pi\epsilon}. \tag{14.2}$$

An independently certified pre-projection HS perturbation ξ may be added to the right side if the perturbed matrix is self-adjoint and recentered in the same physical space.

Proof. The cap difference is $(2\pi)^{-1}\log(\epsilon/d)_+$. Layer cake gives for every x

$$\int \log(\epsilon/d(x,y))_+^2 \,\mathrm{d}\mu(y) = 2\int_0^\infty t\mu(B_d(x,\epsilon e^{-t}))\,\mathrm{d}t \le \frac{2C\epsilon^\alpha}{\alpha^2}.$$

Integrate in x and divide by $4\pi^2$. The resulting HS norm is at most the first term of (14.2). Next $|d(x,y) - d(x_a,x_b)| \le 2h$, and $s \mapsto \log\max\{s,\epsilon\}$ is globally $1/\epsilon$-Lipschitz on $[0,\infty)$. Thus $|G_\epsilon(x,y) - \widehat{G}(x,y)| \le h/(\pi\epsilon)$ pointwise. Since μ is a probability, this also bounds its $L^2(\mu^2)$ norm. Centering contracts HS norm, so the triangle inequality gives the bound for $\|K - \widehat{K}\|_{\mathrm{HS}}$.

Write $\widehat{K} = S - N$ with $S, N \ge 0$ and $SN = 0$. Then

$$\|K - \widehat{K}\|_{\mathrm{HS}}^2 = \|K - S\|_{\mathrm{HS}}^2 + \|N\|_{\mathrm{HS}}^2 + 2\,\mathrm{Tr}(KN) \ge \|K - S\|_{\mathrm{HS}}^2.$$

The trace is nonnegative because K, N are positive and N has finite rank. Constants are in the zero eigenspace of $\widehat{K}$, hence remain annihilated by its positive part. Its range is in the centered cell-constant space, of dimension $q-1$. The same argument after a self-adjoint centered perturbation proves the final assertion. □

14.1 The actual finite matrix

In the orthonormal cell basis $e_a = \mathbf{1}_{C_a}/\sqrt{w_a}$, let $v = (\sqrt{w_a})_a$, $\|v\| = 1$, and put

$$M_{ab} = \sqrt{w_a w_b}\, G_\epsilon(x_a,x_b), \quad Q_c = I - vv^{\mathsf{T}}, \quad \widehat{M} = Q_c M Q_c, \quad S_c = (\widehat{M})_+. \tag{14.3}$$

Lifting S_c by the orthonormal synthesis gives exactly S. In raw indicator coordinates the Gram matrix is $\mathrm{diag}(w_a)$; taking a Euclidean positive part before the physical similarity transform is not justified. Zero-mass cells are omitted. The partition is finite and does not replace μ by an atomic measure in the error theorem: its matrix is lifted to the original $L^2(\mu)$ space as a piecewise-constant kernel.

For nonuniform cell diameters h_a, the same proof permits replacing h by the mass-weighted value $h_{\mathrm{rms}} = (\sum_a w_a h_a^2)^{1/2}$. Indeed the entry error is at most $(h_a + h_b)/(2\pi\epsilon)$, so its squared HS norm is at most

$$\frac{2\sum_a w_a h_a^2 + 2(\sum_a w_a h_a)^2}{4\pi^2\epsilon^2} \le \frac{h_{\mathrm{rms}}^2}{\pi^2\epsilon^2}.$$

This proves an adaptive mass-aware error certificate, without asserting an adaptive complexity advantage. Exact physical masses remain required.

14.2 Rate and its honest interpretation

Put $a = \alpha/2$, $A = \sqrt{C}/(\sqrt{2}\pi\alpha)$, and $D = 1/\pi$. When $\epsilon_* = (Dh/(aA))^{1/(a+1)} \le r_0$, differentiation gives

$$\eta(h, \epsilon_*) = (1+a)A^{1/(a+1)}(Dh/a)^{a/(a+1)}. \tag{14.4}$$

For $\alpha = 1/56$ this is $O(h^{1/113})$. This is deliberately conservative and extremely slow. It is a deterministic continuum-kernel mesh estimate, not an error in n, and not a claim of computational efficiency. Actual numerical certification requires numerical upper bounds for C, a valid r_0, geometric diameter bounds, and exact or certified cell masses. The source's almost-sure existence of random constants does not supply those data. Unknown mass errors cannot simply be labelled a known ξ.

For normalized round area, $\mu(B_d(x,r)) = r^2/4$ for $0 \le r \le 2$. The uncentered cap error is exactly

$$\|G - G_\epsilon\|^2_{L^2(\mu^2)} = \frac{1}{4\pi^2}\int_0^\epsilon \log(\epsilon/r)^2 \frac{r}{2}\,\mathrm{d}r = \frac{\epsilon^2}{32\pi^2}.$$

It attains the general cap bound for $\alpha = 2, C = 1/4$. Centering can improve it, but equality is only claimed before centering.

15 Positive HS inverses control heat traces

The next local theorem is the parent audit's proposed improvement to the *final* heat step. The infinite-dimensional eigenvalue inequality is proved here so that the constant and norm hypotheses are explicit.

Lemma 15.1 (Positive HS eigenvalue perturbation)**.** *Let K, S be nonnegative self-adjoint HS operators on the same separable Hilbert space. List their eigenvalues decreasingly, appending zeros after finite rank, as k_i, s_i. Then*

$$\sum_i (k_i - s_i)^2 \le \|K - S\|^2_{\mathrm{HS}}. \tag{15.1}$$

Proof. For finite spectral truncations, the squared overlaps of the two orthonormal eigenvector families form a sub-doubly-stochastic matrix a_{ij}. For decreasing nonnegative finite sequences k_i, s_j, $\sum_{ij} a_{ij} k_i s_j \le \sum_i k_i s_i$. One direct proof writes $k_i = \sum_{r \ge i}(k_r - k_{r+1})$ and similarly for s_j; each rectangle satisfies $\sum_{i \le r, j \le q} a_{ij} \le \min(r, q)$ by the row and column sum bounds, and the identity pairing attains these bounds simultaneously. This gives $\mathrm{Tr}(K_N S_N) \le \sum_{i \le N} k_i s_i$ for equal-sized truncations. Their HS convergence and Cauchy–Schwarz pass to $\mathrm{Tr}(KS) \le \sum_i k_i s_i$. Expand $\|K - S\|^2_{\mathrm{HS}} = \sum_i k_i^2 + \sum_i s_i^2 - 2\,\mathrm{Tr}(KS)$ to conclude. This is the positive compact HS form of the classical Hoffman–Wielandt principle; the proof needs no commutativity. □

Define for $t > 0$

$$f_t(x) = \begin{cases} e^{-t/(2x)}, & x > 0, \\ 0, & x = 0, \end{cases} \qquad H_K(t) = 1 + \mathrm{Tr}\, f_t(K). \tag{15.2}$$

The $+1$ is one physical constant mode, independent of the possibly infinite-dimensional zero space of a finite-rank inverse extension.

Theorem 15.2 (Uniform quantitative heat transfer)**.** *Let $K, S \geq 0$ be self-adjoint HS and $\eta \geq \|K - S\|_{\mathrm{HS}}$. For every $\delta > 0$ their heat traces are finite and*

$$\sup_{t \geq \delta} |H_K(t) - H_S(t)| \leq \frac{54e^{-3}}{\delta^2} \eta(\|K\|_{\mathrm{HS}} + \|S\|_{\mathrm{HS}}) \tag{15.3}$$
$$\leq \frac{54e^{-3}}{\delta^2} \eta(2\|S\|_{\mathrm{HS}} + \eta).$$

If $\operatorname{rank} S \leq r$*, also*

$$\sup_{t \geq \delta} |H_K(t) - H_S(t)| \leq \frac{8e^{-2}}{\delta} \sqrt{r}\, \eta + \frac{16e^{-2}}{\delta^2} \eta^2. \tag{15.4}$$

Both bounds are valid simultaneously; their minimum may be used.

Proof. Put $y = t/(2x)$. The scalar bound

$$f_t(x) \leq 16e^{-2} t^{-2} x^2$$

follows from $\sup_{y \geq 0} y^2 e^{-y} = 4e^{-2}$. Thus $f_t(K)$ is trace class. For $g_t(u) = f_t(\sqrt{u})$, including the continuous extension at zero,

$$g_t'(u) = \frac{t}{4u^{3/2}} e^{-t/(2\sqrt{u})} = \frac{2}{t^2} y^3 e^{-y} \leq \frac{54e^{-3}}{t^2}.$$

It follows that

$$\sum_i |f_t(k_i) - f_t(s_i)| \leq \frac{54e^{-3}}{t^2} \sum_i |k_i - s_i|(k_i + s_i) \leq \frac{54e^{-3}}{t^2} \|k - s\|_2 (\|k\|_2 + \|s\|_2).$$

Lemma 15.1, the HS spectral identities, and $t \geq \delta$ prove (15.3). The triangle inequality gives its computable second line. For rank r, use $\sup_{x \geq 0} f_t'(x) = 8e^{-2}/t$ on the first r terms and Cauchy–Schwarz. The remaining terms have $s_i = 0$ and $\sum_{i>r} k_i^2 \leq \sum_i (k_i - s_i)^2 \leq \eta^2$. Apply the preceding quadratic bound on f_t to this tail. This also covers $r = 0$. □

Corollary 15.3 (Source heat limit with a shorter final input)**.** *If the source's positive centered inverses converge in HS on the common tour space, then their heat traces converge uniformly on every $[\delta, \infty)$, $\delta > 0$. No new polynomial tail assumption is needed at this final step.*

Proof. HS convergence bounds both HS norms and makes $\eta_n \to 0$ in Theorem 15.2. □

The source's $p > 1/2$ remains a substantive upstream sufficient condition for proving HS convergence in Theorem 10.2. This corollary does not remove it there. It provides no bound at $t = 0$, no pointwise heat-kernel convergence, and no FK convergence rate without an independently proved η_n.

16 Eigenvalue enclosures, padding, and an assembled certificate

For positive compact operators, min–max gives $|k_i - s_i| \leq \|K - S\|_{\mathrm{op}} \leq \eta$. If K is the centered inverse of the raw continuum generator and $s_i > \eta$, reciprocation gives the useful ordered enclosure

$$\frac{1}{2(s_i + \eta)} \leq \Lambda_i \leq \frac{1}{2(s_i - \eta)}. \tag{16.1}$$

No upper bound follows from this formula when $s_i \le \eta$; that is a resolution limitation, not proof of an extra zero eigenvalue. Near multiplicities the ordered values are controlled, but individual eigenvectors are not canonically identified. Stable spectral subspaces require a gap and a separate projection estimate.

For finite graphs, $+\infty$ padding of the generator spectrum corresponds exactly to zero padding of the inverse spectrum. A fixed positive continuum inverse eigenvalue eventually lies above the approximation error, so the corresponding graph index eventually exists. The constant mode remains zero in the generators and contributes exactly one in heat traces.

Theorem 16.1 (Conditional measure-aware Ram certificate)**.** *Suppose* (6.1) *holds, its chordal upper version* (14.1) *has known valid constants, and the partition data in Theorem 14.1 are certified in the physical measure. Let S be the positive centered matrix lift in* (14.3)*, with separately bounded pre-projection error ξ. Then with $\eta = \eta(h, \epsilon) + \xi$:*

1. *The global Green equation and its chartwise Ram integrals* (11.2) *are exactly the same operator equation.*
2. *$\|K - S\|_{\mathrm{HS}} \le \eta$, and all resolvable ordered eigenvalues obey* (16.1)*, including their multiplicities.*
3. *The heat trace computed as $1 + \sum_{s_i > 0} e^{-t/(2s_i)}$ obeys both* (15.3) *and* (15.4) *with $r \le q - 1$.*

If instead a certified full four-point $L^2(\mu^4)$ error η_4 is available for centered kernels, one may use $\eta = \eta_4/2$.

Proof. Apply Theorem 11.1, Theorem 14.1, the ordered min–max bound, and Theorem 15.2 in that order. The last assertion is Lemma 10.1. All operators act on the same physical space; no comparison between an unweighted Euclidean matrix and a quantum operator is inserted. □

The theorem is constructive conditional analysis: it specifies finite inputs and exactly what their certified errors imply. It is not a claim that the needed mass oracle, ball constants, or L^2 voltage error are already available for an FK realization. A few sampled voltage quadruples do not establish η_4. Comparing a finite FK operator with this continuum cell operator additionally requires an error for the graph-to- continuum identification and a common-space coupling. Those are not given by matrix precision.

17 Stepwise source and private-corpus correspondence

The following correspondence records which part of Ram Calculus is actually used, rather than merely observing that both discussions involve matrices.

Source step	Private passage	Report realization and boundary
08: trace into $L^2(\mu)$	Physical Gram definition	Use $\int \psi_i \psi_j \, \mathrm{d}\mu$, never area or counting mass by default. The trace theorem remains independent input to valid form coordinates.
08: centered Green inverse	TR117 RLT/IRLT equivalence	Push μ under $y \mapsto x - y$; kernel relabeling alone omits the displacement measure.

Source step	Private passage	Report realization and boundary
09: physical inverse comparison	Metric symmetry and residual decomposition	Use K and $G_r^{-1/2} C_r G_r^{-1/2}$, with leakage and complementary HS blocks. Domain issues prevent generic smooth-generator residuals.
09: energy compactness	Spectral limits: strong versus norm	The all-level bound and converging ranges supply compactness; finite rank or a core property alone does not supply a uniform error.
09: inverse convergence	Domain-aware approximate inversion	A finite inverse needs a complementary-space argument. Here HS kernel control supplies one for the bounded inverse, not for an unspecified unbounded generator.
Positive-time heat trace	Full residual and stability principle	Theorems 15.2 and 16.1 turn an explicit full HS budget into a readout error, with correct constants.
Local kernel approximation	TR117 exactness and remainder discussion	Capping and mass-weighted cell approximation avoid unsupported Taylor jets and quantify the omitted singular piece.

17.1 Attribution of the local additions

The global/Ram kernel correspondence is attributed to the private TR117 report. Physical Gram transport and the orthogonal residual identity are attributed to the two volume 21 chapters. Their specialization to the singular sphere and the explicit pushforward-measure formulation are derived here. The source's trace and inverse compactness arguments are reconstructed with attribution to M376, not presented as newly discovered theorems.

The parent independently supplied the clock/heat proof note and the capped- kernel certificate note during this investigation. Their statements are proved in this report and their frozen audit files are referenced in the evidence manifest. The smooth-generator-domain proposition is a local deduction under mutual singularity; it is not claimed as a global novelty. All proofs use standard Hilbert-space, capacity, and potential-theoretic tools with explicit hypotheses. The term "Ram" identifies the localized representation and its metric-aware use, not a replacement for those tools.

18 Checks and evidence

Two kinds of verification accompany the paper. Proofs establish the infinite-dimensional statements under their assumptions. Deterministic checks verify algebra, constants, and finite implementations; they do not sample or prove random FK asymptotics.

The independent parent checker reports 414 connected multigraphs with loops and multiplicities, 4,140 exact rational assertions, and 414 weighted four-point identities. It also reports 400 noncommuting positive-semidefinite matrix pairs of dimensions 1 through 10, with 8,800 floating regression assertions, including 2,800 time tests. The maximum observed actual error divided by the minimum

of the two heat bounds is approximately 0.9996157934428739. The numerical part is regression evidence, not interval arithmetic. These parent artifacts were read but neither rerun nor modified.

The author's separate deterministic checker covers the rational GMC exponents, the all-level exponent ledger, the derivative maxima underlying the heat constants, the exact uniform-sphere cap integral, the optimized 1/113 exponent, a singular discrete-measure Ram change of variables, and finite positive-part and block-residual identities. Its JSON output records exact versus floating checks and fixed seed information. The completed author run passed 16 exact symbolic checks and 660 floating assertions on 220 positive-matrix cases; the floating checks use stated tolerances, not certified interval arithmetic. The PDF build checks separately record page count, undefined references, overfull boxes, and rendered page contacts; mathematical checks are not substitutes for page inspection.

18.1 What the checks intentionally do not claim

No FK map simulation is offered as validation of conductivity one. No Monte Carlo quantum cell masses are described as exact. No Lean or other formal verification of M376 was run. The source companions' conditional-law, address, and geometric estimates were not independently machine-certified. The finite graph tests can catch a missing factor of two, a dropped loop degree, or incorrect centering, but cannot validate the random limit law.

19 Advantages, limitations, and future work

19.1 Advantages within proved scope

The physical metric keeps all mass conventions visible, and the bounded inverse avoids inappropriate generator-domain assumptions. Measure-aware Ram coordinates are exact even for atomic approximations and singular continuum mass. The capped-kernel construction supplies a full-space error rather than only a projected residual. Positivity repair in physical coordinates preserves the error guarantee. The heat-transfer estimate uses the actual HS error and works uniformly for all $t \geq \delta$, without repeating the polynomial tail argument at the final step.

19.2 Costs and limitations

The conservative $h^{1/113}$ rate is too slow to justify an efficiency claim, and no optimized partition complexity is proved. Quantum cell masses and ball constants may be much harder to certify than the finite matrix spectrum. Smooth form approximations generally do not give smooth generator vectors. Small inverse eigenvalues cannot be safely reciprocated without an error margin. The centered Green kernel remains globally coupled even in displacement coordinates. Uniform control as $t \downarrow 0$ is absent, and annealed rates would need integrability of the random constants. These restrictions are part of the result, not afterthoughts.

19.3 A carefully separated Weyl-law comparison

Berestycki–Wong [13], Theorems 1.1–1.2, proves for its bounded simply connected planar domain with Dirichlet GFF that

$$\frac{N_\gamma(\lambda)}{\lambda} \longrightarrow \frac{\mu_\gamma(D)}{\pi(2-\gamma^2/2)} \quad \text{in probability.}$$

At $\gamma = \sqrt{3}$ its constant is $2/\pi$ in those conventions. This is a comparison target, not imported here as an ordinary-quantum-sphere theorem or an almost-sure uniform bound over discrete FK indices. To use it quantitatively would require matching boundary conditions, sphere law, area normalization, Brownian clock, convergence mode, and small-time estimates. Fixed-positive-time trace convergence does not justify interchanging the $n \to \infty$ and $t \downarrow 0$ limits.

Promising next work is therefore specific: certify actual mass and ball data on controlled continuum approximations; optimize adaptive partitions using local mass information; quantify the graph-to-continuum HS error; and obtain uniform-integrability estimates for random constants. Independent verification of C/P's protected conditional laws and all-level address estimates remains a separate major task. Equivalent coordinates alone neither discharge it nor solve the global FK scaling problem.

20 Read scope, provenance, and reproducibility

The investigation root is

`C:/Users/mural/OneDrive/Documents/OpenAI-722-math`.

Catalogue scope was the M376 row of `00-Applicability-Catalogue/applicability.json` and the M376 row of `_sources/manuscripts.json`, with snapshot metadata and source hashes. No claim to have reviewed all 722 catalogue manuscripts is made.

The source base is

`_sources/openai-math/preprints/`
`Spectral-convergence-for-critical-FK-Ising-planar-maps-October-5-2026/build`.

Read completely: `paper.tex`; `tex/01-introduction.tex`; `tex/02-model-geometry.tex`; `tex/03-locality.tex`; `tex/04-network.tex`; `tex/05-compactness.tex`; `tex/06-conductivity.tex`; `tex/07-green.tex`; `tex/08-continuum.tex`; `tex/09-spectrum.tex`; `tex/references.tex`; `references.bib`; and `figures/mixed-boundary-duality.tex`. The manuscript README and the repository README were also read. Line counts and hashes are in the machine-readable evidence record. Source originals were not edited.

Direct private reading was complete for

`C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/`
`ch01d-physical-metrics.tex`

`C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/`
`ch01f-spectral-limits.tex`.

In

```
C:/Users/mural/OneDrive/Documents/RamFoundationTRs/
TR-rev-1-old-tr117/TR-rev-1-old-tr117.tex
```

the direct scope is lines 238–315 (basic Ram transform, RLT/IRLT, equivalence proof) and 477–785 (linear scheme, remainders, domain, derivative and inversion cautions). Other sections of that report were located by headings, not claimed as fully read. The wider private corpus and the complete proofs of C/P were not read by this author.

The two parent proof notes and the parent JSON results were read in full. Their checker source and any other parent integration files were not independently re-audited by this author. The SHA256 manifest identifies which parent artifacts were referenced without asserting ownership of them. All new report files are in the M376 report directory; author audit files carry the M376-author name; builds are confined to the report-named build directory. Catalogue, progress, queue, and shared helper files were untouched.

20.1 Compiler and visual audit contract

The built-in LaTeX editor and compiler were attempted with waits bounded at 15 and 30 seconds, respectively. The editor request returned queued; the native compiler returned "Unable to find standard directories for platform." Native compilation is therefore not verified. The on-disk PDF is separately exported using the pre-existing helper `_tools/compile_report.py`, installed MiKTeX, and installed Poppler, with no installation. At most two formatting audit rounds are permitted. Each round may contain three TeX passes for cross-references. The delivered audit record reports the actual outcome, page count, and inspection of all rendered pages, with detailed page inspection where necessary. The final source, PDF, checks, and audit evidence are frozen by a SHA256 manifest after verification.

21 Concluding assessment

Within the read scope, the trace, centering, inverse compactness, and final spectral passage form a coherent deterministic chain under their stated inputs. No specific error was identified in those scoped arguments. This is not a statement that all probabilistic inputs or the whole public preprint are correct. The most consequential unresolved dependencies are the protected conditional-law machinery, conductivity identification, and the genuinely all-level energy approximation.

The Ram contribution is correspondingly precise. It yields an exact measure-aware local representation and a disciplined finite physical calculus. Combined with a proved full kernel error, it gives ordered spectral enclosures and quantitative heat-trace bounds. The constructive cap certificate and the HS heat modulus are meaningful local improvements to an analysis and verification workflow. Neither changes the quantum surface law nor turns a conditional random-map theorem into a globally verified result.

References

[1] OpenAI. *Spectral convergence for critical FK–Ising planar maps.* October 5, 2026. OpenAI Math Release preprint, M376. Public pin `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Pinned source driver. Local complete source is the substantive reviewed version; public web fetch of the pinned directory was unavailable in this audit.

[2] OpenAI. *Canonical conformal limits of subcritical FK planar maps*. September 24, 2026 preprint, companion C, as referenced by M376. Pinned companion directory. Imported dependency, not independently fully verified here.

[3] OpenAI. *Metric-measure limits of subcritical FK and spanning-tree planar maps*. September 24, 2026 preprint; October 3 revision specified by M376. Companion P. Pinned companion directory. Imported dependency, not independently fully verified here.

[4] Muralidhara Subbarao. Private Ram Foundation report, `TR-rev-1-old-tr117/TR-rev-1-old-tr117.tex`. Basic Ram Transform and Ram Localization; Equivalence of global and Ram forms; General Linear Ram Solution Scheme. Local private source, read October 7, 2026.

[5] Muralidhara Subbarao. Private Ram Calculus Books, volume 21, `chapters/ch01d-physical-metrics.tex`. Physical Gram definition, metric symmetry, full residual decomposition, approximate Gram metrics, and finite measurements. Local source, read in full.

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[13] N. Berestycki and M. D. Wong. *Weyl's law in Liouville quantum gravity*. https://arxiv.org/abs/2307.05407, version 3, May 30, 2024. Primary HTML domain setup and Theorems 1.1–1.2 inspected: https://arxiv.org/html/2307.05407v3. The rendered internal document date differs from the arXiv version date; the version identifier is used here.

[14] Z.-Q. Chen and Y. Yu. *Discrete Approximation to Time-changed Brownian Motions*. September 7, 2026 preprint, version 1. https://arxiv.org/abs/2609.07789. Primary abstract inspected; not treated as a verified replacement for the FK inputs.

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